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Variety Seeking, Purchase Timing, and the “Lightning Bolt” Brand Choice Model

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The “Lightning Bolt” (LB) model provides a comprehensive framework for accommodating the effects of habit persistence, unobserved heterogeneity, and state dependence on household brand choice behavior. This paper presents a discrete, dynamic brand-choice model that belongs to the LB class of models. We propose a method for incorporating the effects of variety seeking into the LB model formulation. The proposed formulation explicitly links brand choice and purchase timing behavior via the effect of state dependence. This state-dependence based linkage between brand choice and purchase timing comes about due to the attribute satiation notion associated with household variety seeking behavior. Empirical implementation of the model specification requires recognizing that interpurchase times, like brand choices, also depend upon marketing variables and household characteristics. A hazard model is specified to capture this relationship.

Empirical results are presented using household-level data for two different products. Our results reveal the following. (1) State dependence effects decline over time. Hence, households’ brand switching and repeat purchase probabilities vary over time, independent of any variation in marketing mix activities. (2) Accounting for the effects of marketing variables on interpurchase times may be important when empirically estimating the proposed model. While the *nature* of substantive implications is largely unchanged, the magnitudes of the different effects are nevertheless affected. (3) We find evidence for purchase reinforcement (and its effects declining over time), but no evidence for attribute satiation based variety seeking behavior. (4) A comparison of the model’s predictive ability with those from two extant specifications reveals that the proposed model outperforms rival specifications. (*Brand Choice Dynamics; Hazard Models; Perceptual Maps*)

Introduction

In a recent paper, Roy, Chintagunta and Haldar (1996), hereafter referred to as RCH, presented an empirical application of the “Lightning Bolt” (LB) model of dynamic consumer brand-choice behavior developed by Roy (1990) and Resnick and Roy (1990). An important feature of this dynamic, reduced-form, discrete-choice model is that it is derived from consumer utility maximizing behavior. The framework accommodates several aspects of choice behavior,

such as habit-persistence, heterogeneity, and purchase reinforcement (see Heckman (1981) for an elaboration on these different aspects). Further, it nests several models presented in the marketing literature such as the logit model of Guadagni and Little (1983), the stochastic brand-choice model in Bass et al. (1976), the heterogeneous logit choice model (Chintagunta et al. 1991), among others. RCH highlight the empirical insights obtained from the LB model and demonstrate its substantive usefulness in separating out the effects

of habit-persistence, heterogeneity, and purchase reinforcement.

A key substantive limitation of the RCH study as noted therein, was the inability to isolate variety seeking effects in the LB model. In other words, while the authors showed empirically that one could uniquely identify heterogeneous preferences, habit persistence effects and the impact of purchase reinforcement, they concluded that the model structure precluded isolating variety seeking behavior (if any) of consumers. Not accommodating variety seeking behavior implied that the LB model was an incomplete abstraction of dynamic brand choice behavior. Substantively, one of the objectives of this study is to show how variety seeking behavior can also be identified within the context of the LB model.

As noted by McAlister and Pessemier (1982), a widely accepted characterization of variety seeking is based on the notion of attribute satiation. The idea here is that by purchasing (and consuming) a brand, the household is "satiated" with that brand's attributes/attribute levels and, therefore, seeks out brands that have attributes/attribute levels very different from those of the brand just purchased. Over time, the satiation diminishes, and as a consequence, so does the variety seeking tendency of the household. How can these effects be incorporated into the LB framework?

Note that there are two features to the above notion of variety seeking. One is that it is *attribute* based. The other is a *temporal* variation in the desire to seek variety. Recently, Erdem (1996) has proposed a method for incorporating the effects of attribute-based variety seeking and purchase reinforcement in logit brand choice models. Specifically, her approach is as follows. Households' preferences for brands can be decomposed into brand locations in attribute space and household-specific importance weights for these attributes. A variety seeking (purchase reinforcement) household has a lower (higher) probability of purchasing a brand whose attribute levels are closest to those for the previously purchased brand. Hence, the Erdem study accounts for the attribute-based nature of variety seeking but does not capture the temporal dimension of this phenomenon.

In this study, we incorporate the spirit of the Erdem

approach into the LB modeling framework. Further, we account for the temporal nature of variety seeking/purchase reinforcement by making the extent to which a household seeks variety along a particular attribute vary with the time elapsed from purchase. In this manner, the effects vary with time, and are specific to the attribute as well as to the household. Hence, we account for this additional source of heterogeneity across households.

It would seem from the above description that one could simply reestimate the LB model used by RCH after allowing for the functional relationship between variety seeking/purchase reinforcement and interpurchase times. However, such estimation raises the following *empirical* issue. We know that marketing variables affect dynamic brand choice (as in the RCH study) as well as purchase timing behavior (as shown by Jain and Vilcassim (1991) and others). Further, according to the proposed approach above, brand choice depends upon purchase timing (due to temporal variety seeking effects), thereby linking these two aspects of purchase behavior (namely, brand choice and purchase timing). Therefore, ignoring the effects of marketing variables on purchase timing while estimating the effects of these variables on brand choice may result in incorrect inferences regarding the effects of marketing variables in the LB model. The point being that appropriate empirical implementation of the LB model requires us to consider both brand choice and purchase timing information and the impact of marketing variables on both these decisions.

Consider the effects of a price cut for brand A on a household that had previously purchased brand A. First, there is the direct effect of the price cut on the transition probabilities of buying the various brands, given the previous purchase of brand A. In addition, such a price cut could potentially accelerate the purchase decision for the household; this would change the level of variety seeking/purchase reinforcement, which in turn affects the transition probabilities. This is the indirect effect that is accounted for by the proposed versions of the LB model. In this illustration, not accounting for the relationship between price and purchase timing will lead to ignoring the indirect effects of price.

It is important to note, however, that the LB model does *not* specify a functional relationship between interpurchase times and the effects of marketing activities of brands. Given this absence of a theoretical model for purchase timing within the context of the LB model, any chosen functional relationship is ad-hoc in its specification. Justification for a specific model will, therefore, have to come from the existing literature and/or be based on empirical criteria such as model fit. The literature in marketing for studying the relationship between the category purchase timing behavior and its determinants recommends use of the hazard function methodology (Jain and Vilcassim 1991). Accordingly, we use the hazard function approach to relate the interpurchase times to marketing and household variables. The choice of the parametric distribution for the interpurchase times is based on which of several "reasonable" specifications (Box-Cox, log-logistic, inverse Gaussian, generalized Gamma) provides the best fit to the data.

So the complete *econometric* specification needed to estimate the parameters of the LB model consists of brand choice probabilities and transition probabilities that are functions of marketing variables and the interpurchase times between category purchases. This would be the contribution of the LB model to the likelihood function. Interpurchase times are explicitly linked to marketing variables and consumer characteristics via a hazard model specification. This would be the contribution of the timing model to the likelihood function. The LB model links the purchase timing and brand choice decisions. This linkage necessitates the simultaneous estimation of the effects of marketing variables on the purchase timing and brand choice decisions.

The contributions of this study are as follows. Substantively, the issues of heterogeneity, habit persistence, purchase reinforcement and variety seeking are all investigated using a single framework. The notion of attribute satiation is also accommodated, *vis-à-vis*, variety seeking behavior. Allowing for the different dynamic aspects of purchase behavior in a single model allows us to isolate each of their effects. From a methodological perspective, we provide a fuller characterization of the LB model that explicitly

attempts to incorporate the effects of variety seeking. This is also the first study that simultaneously examines brand choice and purchase timing behavior using a model in which these two decisions are interdependent. Previous studies, e.g. Gupta (1988), model these decisions independently. Additionally, we extend the Erdem (1996) approach to allow for time-varying variety seeking behavior.

The rest of the paper is organized as follows. The next section provides the model formulation and specification. The following section deals with empirical results using data from a frequently purchased product category. This section also provides a comparison of the results with some alternative specifications. We conclude in the final section.

Model Specification

According to the LB model specification (Roy et al. 1996), the brand choice probability of choosing brand m ($m = 1, 2, \dots, M$) at time t , $P[J_t = m]$ reduces to the logit formulation as follows:

$$P[J_t = m] = \frac{\exp(v_t^m)}{\sum_{i=1}^M \exp(v_t^i)},$$

where v_t^m is the observed component of brand m 's signal. And the transition probability of choosing brand r at time s and brand m at time t , where $0 < s < t$ (and $\tau = t - s$ is the interpurchase time) is given by:

$$P[J_t = m | J_s = r] = (1 - \rho) \frac{\exp(v_t^m)}{\sum_{i=1}^M \exp(v_t^i)}, \quad \text{if } m \neq r,$$

$$P[J_t = m | J_s = r] = (1 - \rho) \frac{\exp(v_t^m)}{\sum_{i=1}^M \exp(v_t^i)} + \rho,$$

if $m = r$,

where $0 < \rho < 1$ is an unknown parameter and is referred to as *habit persistence*. A comparison with the inertia model of Bass et al. (1976) immediately reveals that it is analogous to their *inertia* parameter ρ , albeit at the individual consumer level. In their empirical analysis, Roy et al. (1996) find that for the data set used, ρ is small in magnitude when state dependence effects are accounted for (although habit persistence and state dependence are econometrically identified).

The above discussion brings up the issue of the other sources of choice dynamics, i.e., state dependence and unobserved heterogeneity. Both of these are accounted for via a more complete specification for the v_t^m function. The systematic component of utility for brand m and household i is:

$$v_t^{mi} = \alpha^{mi} + \beta^i X_t^{mi} + \theta^i I_s^{mi},$$

where α^{mi} is the intrinsic preference for brand m ; β^i is the vector of covariate effects and X_t^{mi} is the vector of covariates for brand m ; θ^i is the effect of state dependence, i.e., the influence of the previous purchase on the current occasion, and I_s^m is an indicator variable that takes the value 1 if brand m was purchased on the previous occasion, s and 0 otherwise. If the estimated value of θ^i is greater than zero, the data are consistent with purchase reinforcement, when equal to zero we have no state dependence and when the estimated parameter is less than zero, we have variety seeking.

Under a random effects specification for heterogeneity, the vector of preferences, $\alpha_i = (\alpha^{1i}, \alpha^{2i}, \dots, \alpha^{Mi})$ for household i is assumed to be a draw from some continuous (Erdem 1996) or discrete (Chintagunta et al. 1991) distribution across households. As the number of brands in the analysis increases, estimating the parameters of this distribution becomes a cumbersome task. Several researchers (Chintagunta 1994, Erdem 1996) have suggested imposing a factor structure on the distribution of preferences. Under such an assumption, we can interpret α_i as a linear function of the brands' time invariant attributes which lie within an F -dimensional map. The vector α_i can then be decomposed as

$$\alpha_i = \mathbf{A} \mathbf{w}_i,$$

where $\mathbf{A}$ is a $M \times F$ matrix of the positions of the M brands along the F attributes and $\mathbf{w}_i$ is an $F \times 1$ vector of household i 's importance weights for these attributes. This formulation assumes that the matrix $\mathbf{A}$ of brand locations is invariant across households whereas $\mathbf{w}_i$ varies across households, following some discrete or continuous distribution. Consequently, α_i , i.e., the brand preferences for household i varies across households. Researchers have interpreted the

brand locations in matrix $\mathbf{A}$ as representing panelists' perceptual map. Hence, a byproduct of this approach is some insight into market structure in the product category of interest (Elrod 1988).

When $F \ll M$, the above decomposition provides a parsimonious representation of the distribution of preferences. The "optimal" F is obtained by estimating the model with an increasing number of attributes until there is no significant improvement in model fit. In most practical situations it has been found that $F = 2$ adequately describes the data. A caveat is that these are unobserved attributes that are open to subjective interpretation after parameter estimation.

The systematic component of brand utility can now be written as:

$$v_t^{mi} = A^m w^i + \beta^i X_t^{mi} + \theta^i I_s^{mi},$$

where A^m is the m th row of the matrix $\mathbf{A}$. Having introduced attributes into the LB model, the next issue is how to make variety seeking and purchase reinforcement attribute specific. This is accomplished by replacing the term $\theta_i I_s^{mi}$ in the above equation by the term

$$\theta_i I_s^{mi} = - \sum_{f=1}^F \theta_{if} (A_f^s - A_f^m)^2,$$

where A_f^m is the location of brand m on attribute f and A_f^s is the location of the brand purchased on the previous occasion, s along attribute f (see Erdem 1996). In other words, state dependence is now characterized by the distance of the brand under consideration (m) from the previously purchased brand. The reinforcement and variety seeking aspects are then accounted for by the household- and attribute-specific θ parameters. Specifically, if $\theta > 0$, then the household has a diminished utility from purchasing a brand that is perceptually distant from that previously purchased, i.e., purchase reinforcement. On the other hand, if $\theta < 0$, then the household derives utility from buying a perceptually distant brand from that purchased previously, i.e., variety seeking. Further, the formulation allows a household to seek variety along one attribute, $\theta_{i1} < 0$, and reinforcement on the other, $\theta_{i2} > 0$.

To account for attribute satiation, one must allow the extent of variety seeking/reinforcement to vary over time. For this, θ_{if} is operationalized as follows: $\theta_{if} = a_{if} + b_f\tau$, where a_{if} and b_f are parameters to be estimated, and τ denotes interpurchase time ($t - s$) as before. Here a_{if} denotes the *time-invariant* component of purchase reinforcement/variety seeking, whereas b_f is the *time varying* component. This formulation can be used to test various hypotheses. For example, if $b_f = 0$ for all f , then state dependence is time invariant. If $a_{if} < 0$ and $b_f > 0$, then we have support for satiation along attribute f . So the final specification for the systematic utility component is:

$$v_t^{mi} = A^m w^i + \beta^i X_t^{mi} - \sum_{f=1}^F (a_{if} + b_f\tau)(A_f^s - A_f^m)^2.$$

The parameters of the above formulation are identified under the following conditions for $F = 2$. (i) One brand lies at the origin of the two-dimensional map (translation). (ii) Another brand lies along one of the attributes (rotation). (iii) w^i is distributed so as to account for scale invariance (see Erdem (1996) when the distribution is continuous and Chintagunta (1994) when the distribution is discrete). Similar conditions can be obtained for $F > 2$. As with the Roy et al. (1996) study, we constrain ρ to be time invariant. Then, one can unambiguously associate the temporal dimension of the LB model with state dependence. The expression for the transition probability is then given by:

$$(1 - \rho) \frac{\exp(A^m w^i + \beta^i X_t^{mi}) - \sum_{f=1}^F (a_{if} + b_f\tau)(A_f^s - A_f^m)^2}{\sum_{l=1}^M \exp(A^l w^i + \beta^l X_t^{li}) - \sum_{f=1}^F (a_{if} + b_f\tau)(A_f^s - A_f^l)^2} + \rho I,$$

$I = 1$ if $m = r$ and 0 otherwise.

The above expression explicitly links the brand choice probabilities with interpurchase times (τ). As noted previously, marketing variables are related to these times. Next we discuss how we account for this relationship.

Purchase Timing

In order to account for the effects of marketing activities and household characteristics on the interpur-

chase times τ , we use a hazard function specification. Specifically, the parametric form employed is that of a log-logistic distribution and an accelerated failure time specification is used. Choice of specification was based on an extensive comparison with Box-Cox, inverse gaussian, and log-normal models, using proportional hazards as well as accelerated failure time formulations.¹ The hazard function $h(\cdot)$, the survivor $S(\cdot)$, and the density $f(\cdot)$ for the log-logistic are given by:

$$h(\tau) = \frac{\kappa \lambda (\lambda \tau)^{\kappa-1}}{1 + (\lambda \tau)^\kappa}; \quad S(\tau) = \frac{1}{1 + (\lambda \tau)^\kappa};$$

$$f(\tau) = h(\tau) * S(\tau),$$

where $\lambda = \psi(X_\tau) \exp(\eta_1)$, $\kappa = \exp(\eta_0)$, where η_0 and η_1 are unknown parameters and $\psi(\cdot)$ is a positive-valued function of marketing variables and household characteristics such as income, family size, and household inventory. As the hazard model is at the level of the category, the influence of marketing variables needs to be operationalized at the category level. There are several possibilities. The straightforward method is to include all variables of all brands and estimate separately the effects of each. Another possibility is to estimate a constrained version in which only variables specific to the brand purchased are included. A third approach is to constrain the total number of parameters while including the information on all variables. To accomplish this, we use an inclusive value variable which is the maximum value of the utility for the category or the natural logarithm of the denominator of the brand choice probability expression, $P[J_t = m]$. Therefore, instead of including the marketing variables of all the brands separately, all these variables are summarized in the inclusive value term. This term can be written as:

$$IV_t = \ln \left(\sum_{l=1}^M \exp(v_t^l) \right)$$

Further, as the $\psi(\cdot)$ function takes on values only in R^+ , it is formulated as an exponential function of the

¹ Details of the empirical comparison of these specifications are available from the author.

inclusive value term and of the household characteristics. This completes the specification of the interpurchase time function. It is then straightforward to write out the likelihood function for the joint brand choice-purchase timing model. Standard likelihood function methods can be employed to maximize the likelihood function. The likelihood function for household i , L_i is given by:

$$L_i = \prod_{m=1}^M (P[J_1^i = m])^{\delta_{m,1}^i} \\
 * \prod_{t=1}^{T_i} \prod_{l=1}^M \prod_{m=1}^M (P[J_{t+1}^i = m | J_t^i = 1 | \tau])^{\delta_{lm,t+1}^i} f^i(\tau).$$

The first term in the above expression corresponds to the brand choice probability on the first observation of the stochastic process. The second probability term represents the transition probabilities on the subsequent purchase occasions, *conditional on interpurchase times*. The last term, f^i , denotes the marginal density associated with the hazard function at the time of purchase. This therefore, represents the contribution of the interpurchase time model to the likelihood function. *Note that the transition probabilities in the above equation are functions of the interpurchase times*. Consequently, the latter two pieces of the likelihood function depend on the interpurchase times. The density for the log-logistic model is given by:

$$f^i(t) = \frac{\exp(\gamma_0 + \gamma_1)\psi_t^i[\psi_t^i \exp(\gamma_1)t]^{\exp(\gamma_0)-1}}{[1 + [\psi_t^i \exp(\gamma_1)t]^{\exp(\gamma_0)}]^2}.$$

Two things are noteworthy: (i) When accounting for unobserved heterogeneity, the parameters w , β , and a_f are assumed to be household specific. In addition, when the purchase timing decision is considered, the parameters of the density function above are also assumed to be household specific in the estimation. (ii) An empirical issue that arises when estimating dynamic choice models is that of unobserved "initial conditions," i.e., the stochastic process is not observed from its inception. Consequently, the first term on the right hand side of the likelihood function above is not part of the data being used for estimation. Treating the first observation in the observed, left censored, data

series as the beginning of the stochastic process could result in biased parameter estimates (Heckman 1981). While there is no easy solution to the problem, Roy et al. (1996), using the ideas suggested in Heckman (1981), investigated two approaches to addressing this issue. The first approach is to assume that there is no problem of initial conditions. The probabilities on the first purchase occasion are set to the brands' market shares. The alternative approach is to treat the initial probabilities as unknown parameters and then estimate them using the observed data series. Both of these approaches are implemented in the empirical analysis presented in the next section.

Empirical Results

We carry out the empirical analysis on two scanner panel datasets, nondiet soft drinks and liquid laundry detergents. Both procedures to account for the problem of initial conditions were investigated in all cases. It was found that ignoring the left censoring problem and treating the first observation as the start of the stochastic process had no adverse impact on the estimates. Further, the Schwarz criterion revealed that including the additional parameters to estimate the initial probabilities did not improve model fit. While this does not imply the absence of the initial conditions problem, it does indicate that the inclusion of additional parameters does not mitigate the problem in this case. Hence, results discussed here pertain to noninclusion of the additional parameters. Later in this section, we provide a comparison of the results from the proposed specification with those obtained from two dynamic brand choice models proposed in the marketing literature, that account for variety seeking behavior.

Recall that the proposed approach implies a time varying, attribute- and household-specific variety seeking effect. Heterogeneity is accommodated via a random effects approach. The two datasets used for this purpose are *nondiet* soft drinks and liquid laundry detergents. The top six brands were chosen from each category. As the former data are confidential and come from a Nielsen national sample of panelists, we refer to the brands as brands 1 through 6. These brands are unique, e.g., Pepsi-Cola and Coca-Cola and

there are no variants of the brands included in the data (e.g., Cherry Coke). The brands in the laundry detergent category are Tide, EraPlus, Solo, Wisk, Surf, and All. The first three brands are marketed by P&G and the latter three by Lever Brothers. These data are from Sioux Falls, SD and were obtained from the AC Nielsen company. Descriptive statistics for the data are given below. There are 1041 households in the soft drinks data with mean interpurchase time of 14 days and 400 households in the detergent data with a mean interpurchase time of 62 days.

Soft Drinks	Feature	Display	Price (c/unit)	Brand Share	Detergent	Feature	Display	Price (c/Oz.)	Brand Share
1	0.072	0.171	7.662	0.373	Tide	0.100	0.108	5.947	0.264
2	0.046	0.149	6.922	0.287	Wisk	0.105	0.125	4.720	0.265
3	0.028	0.393	6.507	0.119	EraPlus	0.028	0.037	6.061	0.191
4	0.070	0.317	7.183	0.096	Surf	0.043	0.054	5.290	0.153
5	0.085	0.142	3.911	0.059	Solo	0.017	0.022	5.988	0.095
6	0.042	0.302	6.741	0.066	All	0.001	0.001	3.907	0.033

In the exhibit above, a fixed number scales the prices of all the soft drinks across all brands. As our interest is in the implications for purchase reinforcement/variety seeking and habit persistence, we shall focus on these issues in the discussion of results. It is also of some interest to examine how the perceptual locations of brands (i.e., the A matrix) is affected by including purchase timing information. In Tables 1 and 2 we present the results from the soft drinks and detergent product categories respectively. In both cases, three support distributions were found to be adequate to characterize the heterogeneity in the data based on the BIC criterion.

Table 1 for the soft drinks data reveals that the brand locations (i.e., perceptual map) obtained when the effects of marketing variables on purchase timing behavior are ignored are quite different from those obtained with the purchase timing hazard function. To provide a clearer picture of this finding, in Figure 1 we plot the brand locations in two-dimensional space. The figure clearly reveals that the market-structure implications obtained from the two methods are quite different. We computed a congruence measure for the maps (correlation in interbrand distances across the two maps) which is 0.6 in this case. In variance terms,

Table 1 Perceptual Brand Locations, Inertia and Variety Seeking in Soft Drinks Data (Three Support Heterogeneity Distribution)

Parameter	No Purchase Timing Hazard		With Purchase Timing Hazard	
Location	Attribute 1	Attribute 2	Attribute 1	Attribute 2
Brand 1	1.122 (0.114)	0.019 (0.017)	1.383 (0.086)	3.991 (0.075)
Brand 2	0.528 (0.055)	0.304 (0.098)	2.894 (0.129)	0.447 (0.154)
Brand 3	0.454 (0.060)	-0.307 (0.102)	1.515 (0.078)	-0.880 (0.048)
Brand 4	0.476 (0.086)	-0.622 (0.203)	0.632 (0.067)	-1.121 (0.079)
Brand 5	-0.088 (0.034)	0* (0.035)	0.784 (0.035)	0* (0.035)
Brand 6	0* (0.034)	0* (0.035)	0* (0.035)	0* (0.035)
Mean Weight	1.205 (0.089)	1.051 (0.186)	-0.039 (0.015)	0.638 (0.112)
Reinf./VS	Attribute 1	Attribute 2	Attribute 1	Attribute 2
Support 1: a_{1f}	1.620 (0.376)	-0.052 (0.242)	0.032 (0.048)	0.105 (0.015)
Support 2: a_{2f}	1.732 (0.365)	22.464 (11.631)	4.743 (0.130)	4.462 (0.003)
Support 3: a_{3f}	0.444 (0.118)	0.746 (0.511)	0.398 (0.050)	0.089 (0.008)
b_f	-0.850 (0.248)	0.683 (0.639)	-0.041 (0.006)	-0.008 (0.001)
Habit Persistence				
Support 1	0.0		0.34	
Support 2	0.22		0.01	
Support 3	0.09		0.02	
Support Sizes	0.28 (#1); 0.46 (#2); 0.26 (#3)		0.05 (#1); 0.51 (#2); 0.44 (#3)	

this is 36%. So, consistent with the earlier findings from approach 1, inclusion of the hazard function in the analysis affects the model implications.

Turning to the results pertaining to purchase reinforcement and variety seeking, we find that the time-invariant components are consistent with purchase reinforcement but not with variety seeking in all three supports. Taking into account the time-varying component, households corresponding to support 1 in the "with purchase timing hazard" case display variety seeking when the time elapsed from the last purchase increases. Hence, while we do not find attribute saturation, we do find declining purchase reinforcement and increasing variety seeking for some households. Table 1 also reveals that at least one support shows a nonzero level of habit persistence for these data.

We turn next to the results from the liquid laundry detergent data in Table 2. In Figure 2, we plot the perceptual maps under the two conditions (with and without the purchase timing hazard). First, note that

Table 2 Perceptual Brand Locations, Inertia and Variety Seeking in Detergents Data (Three Support Heterogeneity Distribution)

Parameter	No Purchase Timing Hazard		With Purchase Timing Hazard	
Location	Attribute 1	Attribute 2	Attribute 1	Attribute 2
Tide	1.043 (0.130)	1.988 (0.204)	1.314 (0.298)	1.754 (0.252)
Wisk	0.548 (0.114)	1.186 (0.190)	0.351 (0.118)	1.547 (0.194)
Era Plus	1.453 (0.157)	1.082 (0.129)	1.839 (0.397)	0.912 (0.165)
Surf	1.024 (0.141)	0.279 (0.154)	0.964 (0.227)	0.720 (0.095)
Solo	1.880 (0.195)	0* (0.195)	1.448 (0.332)	0* (0.332)
All	0* (0.195)	0* (0.195)	0* (0.332)	0* (0.332)
Mean Weight	1.838 (0.222)	0.906 (0.157)	1.512 (0.311)	0.710 (0.234)
Reinf./VS	Attribute 1	Attribute 2	Attribute 1	Attribute 2
Support 1: a_{1i}	15.209 (2.503)	1.081 (0.302)	4.176 (1.839)	2.688 (0.829)
Support 2: a_{2i}	1.191 (0.250)	0.544 (0.144)	0.601 (0.283)	0.728 (0.238)
Support 3: a_{3i}	14.302 (2.479)	0.456 (0.180)	2.937 (1.373)	0.076 (0.335)
b_i	-0.016 (0.013)	-0.007 (0.005)	0.007 (0.010)	-0.001 (0.008)
Habit Persistence				
Support 1		0.06		0.04
Support 2		0.15		0.09
Support 3		0		0.39
Support Sizes	0.37 (#1); 0.50 (#2); 0.13 (#3) 0.38 (#1); 0.58 (#2); 0.04 (#3)			

the market structure implications seem to make sense. The P&G brands seem to be perceptually "separated" from those marketed by Levers. Comparing the figures in the upper and lower panels, we find that market structure implications are affected by the presence of the hazard function to account for the effects of marketing variables on purchase timing behavior. But the effects do not appear to be large.

What may explain the differences between the soft drinks and the detergent data? Examining the parameters corresponding to the time-varying component of state dependence (i.e., b_i), we note that none of these parameters is significant. This "decouples" the choice probabilities from the hazard functions. Hence, the parameters are not much affected. The reason we do

find some effect is due to the inclusive value variable which appears in the hazard function formulation. This variable "links" the two decisions in the formulation (given no time effects on reinforcement/variety seeking).

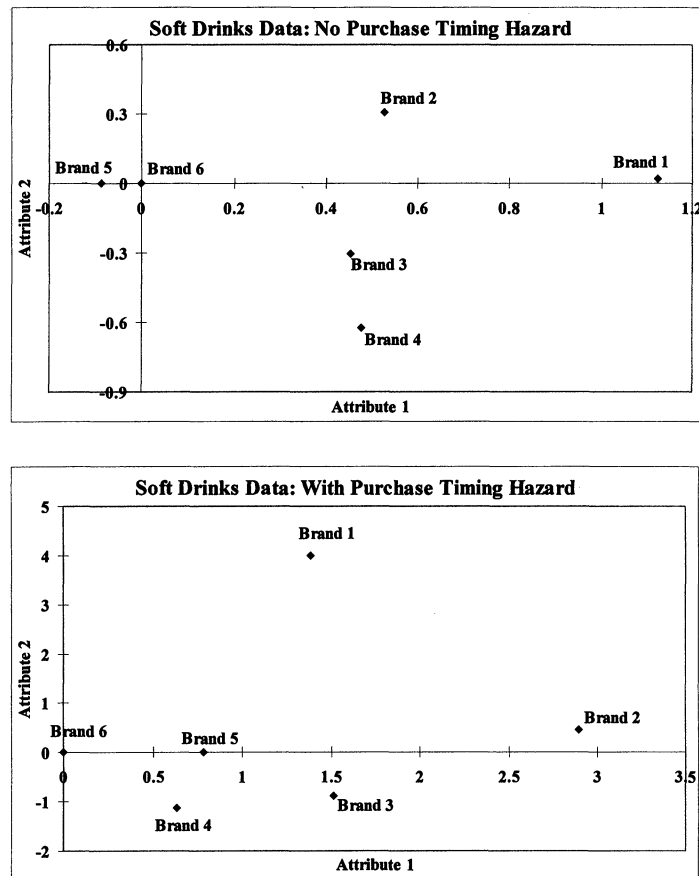
Table 2 also reveals that the detergent data are characterized only by purchase reinforcement and not by variety seeking. This is inferred from the positive values for all the time-invariant, state-dependence parameters. Once again, as with the soft drinks data, we find the presence of some habit persistence even after accounting for unobserved heterogeneity and state dependence. To summarize, while there exist differences across product categories in terms of the extent to which noninclusion of the purchase timing hazard affects results, we do find that it is useful to account for time-varying state-dependence levels.

Having discussed the implications for market structure and the nature of state dependence, we next turn to the estimated heterogeneity distributions. In the interest of space, we focus on the detergent product category. Table 3 provides the relevant parameter estimates and their standard errors.

Table 3 reveals the following. Accounting for the effects of marketing variables on both purchase decisions does seem to impact the estimated effects. Specifically, the display and feature effects with and without accounting for purchase timing seem to differ from one another. While the effects are not as dramatic in the case of price, it is clear that the two distributions are not identical. Additionally, we find a difference in the "importance" weights attached to the two dimensions under the "no purchase timing" and "with purchase timing" scenarios. This implies that the estimated preference distribution across households is affected when purchase timing effects are not accounted for. To obtain more insight into the purchase timing decision, we provide in Figure 3 the estimated baseline hazards corresponding to the three supports for the detergent category.

Figure 3 indicates that the nature of purchase timing behavior is consistent across households. This is revealed by the nonmonotonic pattern exhibited by households corresponding to all three supports of the heterogeneity distribution. This finding is also

Figure 1 Perceptual Maps from Soft Drinks Data



consistent with previous studies using similar data (see for example, Jain and Vilcassim (1991)). There are, however, differences in the baseline hazard function across the three supports. Specifically, the time at which households are most likely to buy varies across households, even after accounting for the effects of marketing and other variables. We find that the hazard peaks at 49 days for support 1, 37 days for support 2, and 15 days for support 3. This indicates that support 3 households are more frequent in their purchases than other households. Overall, the results imply that households differ in their intrinsic purchase timing rates.

Model Comparison

We compared the prediction results from the model with those from two other specifications:

1. The Givon (1984) approach of uncovering inertia and variety seeking from a model whose structure bears some resemblance to the LB model. Specifically, the transition probabilities in the Givon model are given by:

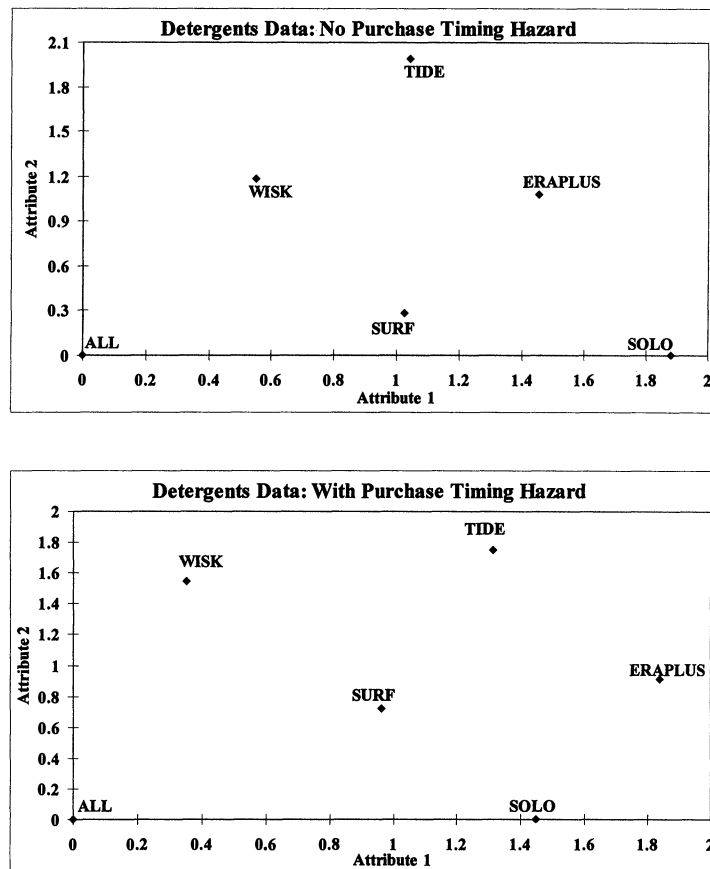
$$P[J_t = m | J_s = r] = \frac{|VS| + VS}{2(M-1)} + (1 - |VS|) \frac{\exp(v_t^m)}{\sum_{l=1}^M \exp(v_t^l)},$$

if $m \neq r$,

$$P[J_t = m | J_s = r] = \frac{|VS| - VS}{2} + (1 - |VS|) \frac{\exp(v_t^m)}{\sum_{l=1}^M \exp(v_t^l)},$$

if $m = r$,

Figure 2 Perceptual Maps from Liquid Laundry Detergents Data



where we have incorporated marketing variables into Givon's original probability expression and $-1 < VS < 1$ captures both inertia and variety seeking. $-1 < VS < 0$ denotes inertia and $0 < VS < 1$ denotes variety seeking.

- The Erdem (1996) approach of incorporating reinforcement and variety seeking effects into the state-dependence term of the conventional logit brand choice model. Here the deterministic component of brand utility for household i and brand m is given by:

$$v_i^{mi} = A^m w^i + \beta^i X_i^{mi} - \sum_{f=1}^F a_{if} (A_f^s - A_f^m)^2.$$

In both cases, we used a random effects approach to account for unobserved heterogeneity.

The validation exercise was carried out on both the soft drinks and detergents product categories. Note that for purposes of comparability across models, we use the results from our model under the "no purchase timing hazard" condition and a random effects heterogeneity specification. In the interest of space we do not provide the detailed results here. The summary of the results is as follows. In both cases, the proposed formulations outperformed the Givon (1984) model on the criterion of predictive log-likelihood. The proposed model also outperformed the Erdem specification for the soft drinks data, but not for the laundry detergents data set. A plausible explanation for this finding is that the time varying component of variety seeking for these data was zero on both attributes. Consequently, with fewer parameters, the

Table 3 Heterogeneity Distribution: Detergents Data

	No Purchase Timing Hazard			With Purchase Timing Hazard		
	Support 1	Support 2	Support 3	Support 1	Support 2	Support 3
Price	-0.890 (0.159)	-1.550 (0.087)	-2.166 (0.274)	-0.820 (0.113)	-1.700 (0.089)	-1.239 (0.227)
Display	0.539 (0.650)	0.753 (0.236)	-0.665 (1.232)	0.089 (0.438)	0.655 (0.239)	2.234 (0.903)
Feature	1.033 (0.612)	1.500 (0.218)	2.792 (1.218)	1.005 (0.415)	1.521 (0.219)	-0.622 (0.846)
w_1	1.000*	1.973 (0.270)	3.833 (0.809)	1.000*	1.859 (0.445)	1.253 (0.509)
w_2	0.544 (0.155)	1.000*	1.619 (0.312)	0.127 (0.145)	1.000*	2.365 (0.508)

* Constrained to 1 for identification purposes.

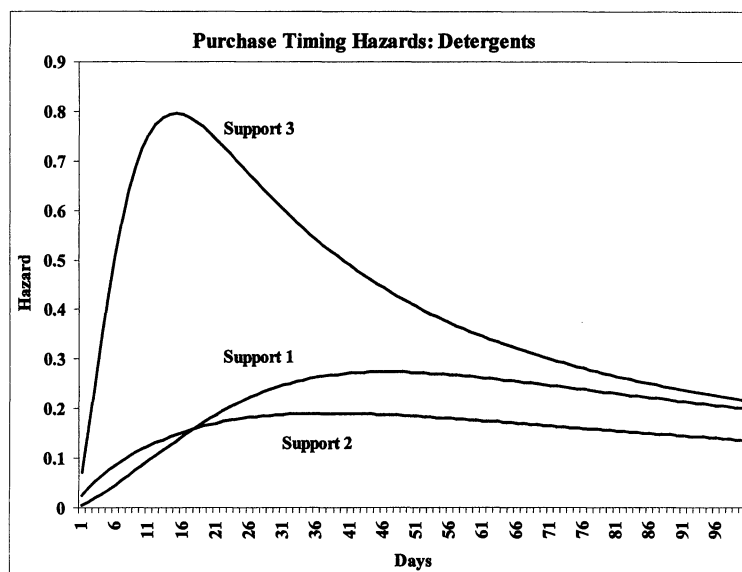
Erdem model captured the variation in the data better than the proposed model.

Managerial Implications

1. The model formulation enables one to partition out the effects of consumer purchase dynamics (i.e., inertia and variety seeking) from the effects of marketing variables and unobserved heterogeneity. In the process, a clearer understanding of both brand choice as well as purchase timing behavior can be obtained.

2. By treating the level of a household's inertia/variety seeking as being time variant, several implications emerge. Consider first, the case reflected in our empirical data, i.e., that households are inertial, but the inertia wears out over time. Suppose there is a detergent in the market that offers its brand in two sizes—small and large—with the larger pack costing less per oz.. Now consider two households (with equal number of members)—one that buys the small size and the

Figure 3 Baseline Hazard Functions for Liquid Laundry Detergents Data



other the larger size. Let us assume that the consumption rate is the same for both households. All else being equal, the household buying the larger size would make its next detergent purchase after the other household. This delay in purchase could potentially be detrimental to the brand, as the level of inertia would have declined a lot more for that household at the time of next purchase. Hence that household is more likely to switch on the next occasion. (It is important to note, however, that a larger size purchase "loads up" the household and precludes the ability to switch to a competing brand). Next, consider a variety seeking household subject to attribute satiation. In this case, purchase of the larger size would benefit the brand, as the desire to seek variety diminishes over time. Consequently, once the household feels the need to purchase detergents, a low level of variety seeking would aid brand repurchase. By explicitly incorporating the temporal dimension of inertia/variety seeking, the proposed model allows us to evaluate these different scenarios.

3. In addition to the above implications, the proposed formulation also allows us to recover market structure. This means that the nature of interbrand competition can be analyzed by exploiting the information contained in brand purchases over time.
4. By exploiting the information contained in the brand choice as well as the purchase timing decisions of households, the model provides a more comprehensive analysis of purchase behavior.

Conclusions

This paper presents a discrete, dynamic brand choice model that belongs to the class of models developed by Roy (1990). We propose a method for incorporating the effects of variety seeking into the model formulation. Further, the proposed formulation explicitly links brand choice and purchase timing behavior via state dependence. As state dependence is an aspect of dynamic choice behavior, the linkage with timing appears to be intuitively plausible. Empirical estimation of the model specification requires recognizing

that interpurchase times depend upon marketing variables and household characteristics. A hazard model is specified to capture this relationship.

Empirical results are presented for two different product categories. The key empirical findings are as follows:

1. State dependence effects decline over time when habit persistence effects are held constant.
2. Accounting for the impact of interpurchase time on purchase reinforcement/variety seeking results in improved model fit.
3. Accounting for the effects of marketing variables on interpurchase times may be important when empirically estimating the proposed model. While the *nature* of substantive implications are largely unchanged, the magnitudes of the different effects are nevertheless affected. This is reflected in the market structure implications obtained.
4. We find evidence for purchase reinforcement (and its effects declining over time), but no evidence for attribute satiation based variety seeking behavior. Further research is necessary to determine whether this can be attributed to the model formulation, to the heterogeneity specification, or to other reasons not considered here.

There are model limitations that provide caveats to the results presented above. First, the model is in reduced form. Structural model results may have to be compared with those presented here in order to gain a better understanding of empirical differences between models. Second, the chosen model specification is based on a single member of the multivariate extreme value family. Some investigation into alternative parameterizations may be warranted. A conceptual problem that arises is that we are trying to identify variety seeking behavior from household level data. However, if we do find evidence of such behavior, the question arises: Is it true variety seeking or is it spurious variety seeking obtained due to the aggregation of inertial purchases of different household members, where each member is loyal to a different brand? Our model specification cannot distinguish between these two explanations. One way of addressing this problem would be

to focus exclusively on single member households. However, this in turn would raise the issue of representativeness of the sample.

In summary, the LB model is a useful framework for isolating the various aspects of dynamic brand choice behavior. The study also finds that purchase timing behavior seems to be related to brand choice behavior via the effects of state dependence.²

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